Practical 14: Started and Bound Services

Source: MODULE 2.pdf | Source pages: 35–43

Aim

To develop and test an Android application demonstrating started and bound services.

Steps to Perform the Practical

1. Create a project named ServiceDemo and add Start Service, Stop Service and Bind Service buttons.

2. Create MyStartedService.java and implement onStartCommand(), onDestroy() and onBind().

3. Create MyBoundService.java with a LocalBinder and a getMessage() method.

4. Add both service declarations inside the element of AndroidManifest.xml.

5. In MainActivity.java, connect the buttons to startService(), stopService() and bindService().

6. Run the app. Test starting/stopping the service and binding to it. Check the Toast messages.

Result

The application is run and the expected behavior is verified according to the practical instructions and code.

Note: Use the package name and resource names from the source code consistently. If Android Studio reports a missing resource or class, check spelling and ensure all files are in the correct folders.

Services, Notifications and Broadcast Receivers

7. Develop an Android application implementing Started and Bound Services.

A bound service is the server in a client-server interface. It lets components such as activities bind to the service, send requests, receive responses, and perform interprocess communication (IPC). A bound service typically lives only while it serves another application component and does not run in the background indefinitely.

Project Name: ServiceDemo

File Name: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout

   xmlns:android="http://schemas.android.com/apk/res/android"

   android:layout_width="match_parent"

   android:layout_height="match_parent"

   android:orientation="vertical"

   android:gravity="center"

   android:padding="20dp">

   <Button

       android:id="@+id/btnStart"

       android:layout_width="wrap_content"

       android:layout_height="wrap_content"

       android:text="Start Service"/>

   <Button

       android:id="@+id/btnStop"

       android:layout_width="wrap_content"

       android:layout_height="wrap_content"
```

```
        android:text="Stop Service"

        android:layout_marginTop="15dp"/>

    <Button

        android:id="@+id/btnBound"

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:text="Bind Service"

        android:layout_marginTop="15dp"/>

</LinearLayout>
```

Create a Started Service Class.

Right Click: java package→ New → Java Class

Class Name: MyStartedService

File Name: MyStartedService.java

```
package com.example.servicedemo;


import android.app.Service;

import android.content.Intent;

import android.os.IBinder;

import android.widget.Toast;


public class MyStartedService extends Service {


    @Override

    public int onStartCommand(Intent intent, int flags, int startId) {
```

```
        Toast.makeText(
                this,
                "Started Service Running",
                Toast.LENGTH_SHORT
        ).show();

        return START_STICKY;
    }

    @Override
    public void onDestroy() {
        super.onDestroy();

        Toast.makeText(
                this,
                "Started Service Stopped",
                Toast.LENGTH_SHORT
        ).show();
    }

    @Override
    public IBinder onBind(Intent intent) {
        return null;
    }
}
```

Create a Bound Service Class (Similar like above).

Create Java Class: MyBoundService

File Name:MyBoundService

```
package com.example.servicedemo;


import android.app.Service;

import android.content.Intent;

import android.os.Binder;

import android.os.IBinder;


public class MyBoundService extends Service {


   private final IBinder binder = new LocalBinder();


   public class LocalBinder extends Binder {

       public MyBoundService getService() {

           return MyBoundService.this;

       }

   }


   @Override

   public IBinder onBind(Intent intent) {

       return binder;

   }


   public String getMessage() {
```

```
        return "Bound Service Connected";
    }
}
```

File Name: MainActivity.java

```
package com.example.servicedemo;


import android.content.ComponentName;
import android.content.Intent;
import android.content.ServiceConnection;
import android.os.Bundle;
import android.os.IBinder;
import android.widget.Button;
import android.widget.Toast;


import androidx.appcompat.app.AppCompatActivity;


public class MainActivity extends AppCompatActivity {


    Button btnStart, btnStop, btnBound;


    MyBoundService myBoundService;
    boolean isBound = false;


    @Override
```

```
protected void onCreate(Bundle savedInstanceState) {

    super.onCreate(savedInstanceState);

    setContentView(R.layout.activity_main);



    btnStart = findViewById(R.id.btnStart);

    btnStop = findViewById(R.id.btnStop);

    btnBound = findViewById(R.id.btnBound);



    // Start Service

    btnStart.setOnClickListener(v -> {



        Intent intent = new Intent(

                this,

                MyStartedService.class

        );



        startService(intent);



    });



    // Stop Service

    btnStop.setOnClickListener(v -> {



        Intent intent = new Intent(

                this,
```

```
                MyStartedService.class
        );


        stopService(intent);


    });
    // Bind Service
    btnBound.setOnClickListener(v -> {
        Intent intent = new Intent(
                this,
                MyBoundService.class
        );


        bindService(
                intent,
                serviceConnection,
                BIND_AUTO_CREATE
        );
    });
}
// Service Connection
ServiceConnection serviceConnection = new ServiceConnection() {


    @Override
    public void onServiceConnected(
            ComponentName name,
```

```
                IBinder service) {

            MyBoundService.LocalBinder binder =

                    (MyBoundService.LocalBinder) service;

            myBoundService = binder.getService();

            isBound = true;

            Toast.makeText(

                    MainActivity.this,

                    myBoundService.getMessage(),

                    Toast.LENGTH_SHORT

            ).show();

        }



        @Override

        public void onServiceDisconnected(ComponentName name) {

            isBound = false;

        }

    };

    @Override

    protected void onDestroy() {

        super.onDestroy();

        if (isBound) {

            unbindService(serviceConnection);

            isBound = false;

        }

    }

}
```

Add Services in AndroidManifest.xml (in Android Project Panel).

Open:

app→ Manifests → AndroidManifest.xml

Inside: <application>

Add:->

<service android:name=".MyStartedService"/>

<service android:name=".MyBoundService"/>

